

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the section as directed.*

Section-A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $2 \times 10 = 20$

(i) What is the primary purpose of a
computer network ?

- ☒ a) To allow computers to communicate and share resources
 - b) To perform calculations faster
 - c) To store data securely
 - d) To connect computers to the internet
- (ii) In which network topology does each device have a point-to-point connection to every other device?
- a) Star topology
 - ☒ b) Mesh topology
 - c) Ring topology
 - d) Bus topology
- (iii) Which of the following protocols is used for sending email?
- a) FTP
 - b) DHCP

☒ c) SMTP

d) SNMP

(iv) Which layer of the OSI model is responsible for routing packets between devices on different networks?

a) Application layer

b) Transport layer

☒ c) Network layer

d) Data Link layer

(v) In the OSI model, which layer is responsible for error detection and correction?

a) Physical layer

☒ b) Data Link layer

c) Network layer

d) Presentation layer

(vi) In Go-Back-N ARQ, what happens if a single frame is lost or corrupted ?

a) The sender retransmits the lost frame only

☒ b) The sender retransmits all frames starting from the lost frame

c) The receiver requests the retransmission of the lost frame only

d) The connection is terminated

(vii) Which multiplexing technique divides the available bandwidth into multiple frequency bands?

a) TDM

☒ b) FDM

c) CDM

d) SDM

(viii) In circuit switching, what happens before the actual data transfer begins?

- a) Data is segmented into packets
- ☒ b) A dedicated communication path is established
- c) Data is encrypted
- d) Routing information is exchanged

(ix) Which protocol is commonly associated with connectionless datagram switching?

- a) TCP
- ☒ b) UDP
- c) FTP
- d) HTTP

(x) In Stop-and-Wait ARQ, what happens if an acknowledgment (ACK) is not received within a certain timeout period?

- a) The next frame is sent
- b) The sender resends the same frame
- c) The connection is terminated
- d) The sender waits indefinitely

Section-B

Short answer type questions

Answer any four questions of the following :

5×4=20

2. How does mesh topology enhance network reliability and fault tolerance compared to other topologies?
3. Define multiplexing. Describe the frequency division multiplexing ?
4. Define transmission media. Mention the name of types of guided and unguided transmission media.
5. Define switching. Describe the packet switching technique.

6. How does the Transmission Control Protocol (TCP) differ from the User Datagram Protocol (UDP) in terms of reliability.
7. Define network. Explain the types of networks.

Section-C

Long answer type questions

Answer any **two** questions of the following :

$15 \times 2 = 30$

8. How stop and wait ARQ protocol handle following problems :
 - (a) Problem of lost data packet
 - (b) Problem of lost acknowledgement
 - (c) Problem of delayed acknowledgement
9. What are the seven layers of the OSI Reference Model, and what is the primary function of each layer?

10. The message 100111001 is to be transmitted using CRC polynomial $x^3 + x + 1$ to protect it from errors. Calculate the message that should be transmitted.
11. Explain the process of Carrier Sense in the CSMA/CD protocol and why it is important for avoiding data collisions.

_____x_____